

1) Optional: BJT-based Astable Multivibrator

Purpose of the laboratory:

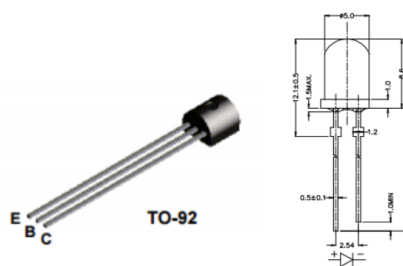
- To study a BJT-based astable multivibrator that alternately turns on and off two LED diodes
- Understand and describe the operation of this circuit

Instrumentation needed:

- Arbitrary waveform generator
- 2-channel oscilloscope
- Power supply
- 1 BNC "T" connector
- 2 BNC male/banana female connectors
- 1 female-to-female BNC connector
- 1 BNC cable
- Cable 1 [mm]
- Wire stripper

Components needed:

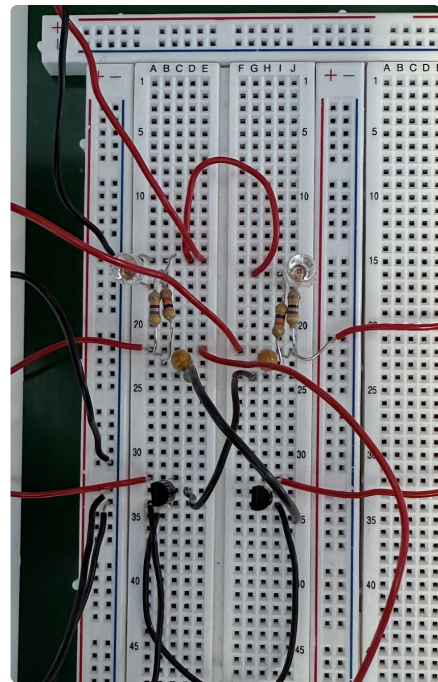
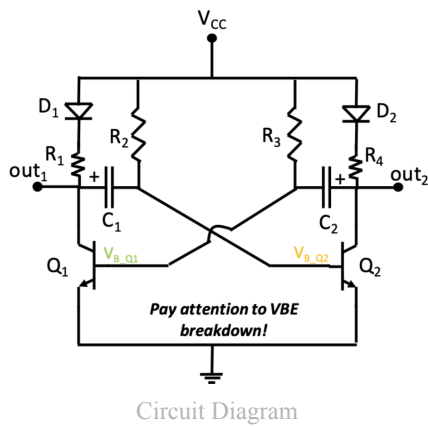
Component Type	Manufacturer Code/Value
2 bipolar npn transistors	BC548
Resistors	R =R14 =470 Ω R =R23 =47 $\kappa\Omega$
Capacitors	C =C12 =10 μF
Red LEDs	CREE C503B-RCS-CW0Z0AA1



BJT and Led Diagram

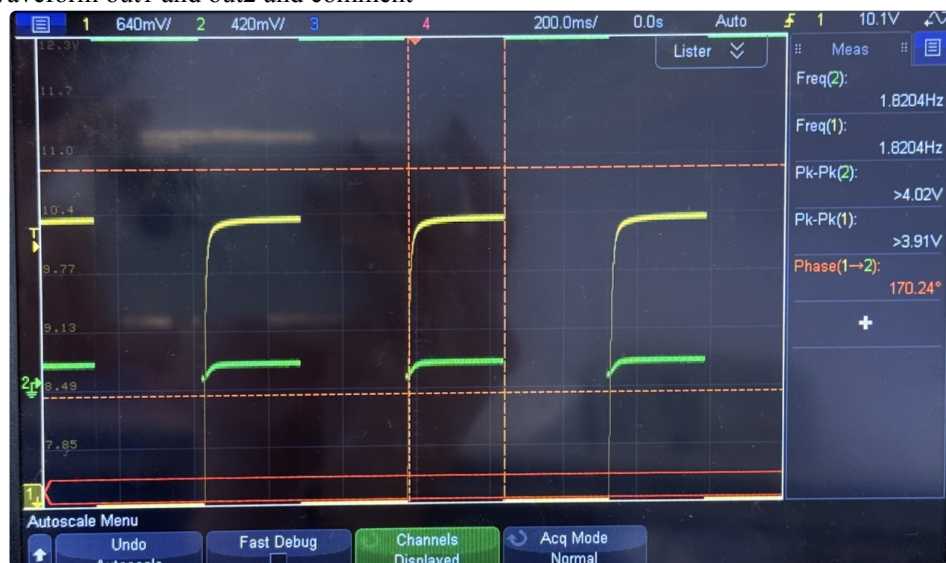
1.1) In the Laboratory

1. Mount the circuit in the figure, and turn on the power supply with VCC =12 V.



Real Life Circuit

2. Measure the waveforms at the base and collector of transistor Q1 and analyze them. Unfortunately this was not done due to lack of time.
3. Report the waveform out1 and out2 and comment



Oscilloscope

Before doing an analysis it is important to describe how the signals are displayed. On a first look it might seem that the two signals are in phase but it is a byproduct of how the reference of the signals were set. Originally, with the same reference they were sovrapposed so by displacing them it was possible to see the alternation. In the yellow signal, what might look as the minimum (near the green signal) is actually the maximum, this can be verified by looking at the phase which is of $170^\circ \approx 180^\circ$. Now it is clear how the circuits work and how the leds were blinking by alternating.

4. Evaluate the period, frequency and duty cycle of the waveform generated at the outputs out1 and out2
In point 3) it was already stated that the two signals were symmetrical (this will be briefly explained later). Since the signal can be approximated in two states 0, V_{CC} the duty cycle can be calculated by comparing the duration of the OFF and ON times. This results in 50%. The period is given by the inverse of the frequency $T = 1/f = 0.54 [s]$. This also equals to the sum of the ON and OFF times.
5. Describe the operation of the circuit
At $t < 0$ both transistors are off. By turning on the circuit and only one of the two will turn on due to small

fabrication differences. Assuming it is Q1, $V_{BE1} = 0.7 [V]$ and therefore $v_{O1} = 0$. Since C_1 is not charged, Q2 is OFF and therefore $v_{O2} = V_{CC}$. C_1 starts to charge and V_{BQ2} will grow. At the same time also C_2 charges until V_{CC} . Once V_{BQ2} reaches $0.7 [V]$ Q2 turns on and $v_{O2} = 0$. As this happens the left side of C_1 goes to $0.7V$ and Q1 turns OFF. Therefore $v_{O1} = V_{CC}$, C_2 starts to change and the process repeats cyclically.

6. How can the period of the waveform be calculated from the values of the resistors and capacitors? Explain in the report how the circuit works

The analytical formula shows that the period is given by the sum of the duration of the two ON states. Ideally, by construction we have a symmetry between the two times since $R_2C_1 = R_3C_2$. By expliciting the formula:

$$T = -\ln \left(\frac{V_{CC} - V_{BE}}{2V_{CC} - V_{BE} - V_{led}} \right) (R_2C_1 + R_3C_2)$$